

Topic 2: Sinusoids

[Summary based on J. H. McClellan, R. W. Schafer, and M. A. Yoder, *Signal Processing First*, Prentice Hall, 2003.]

Topics: Sinusoidal signals, amplitude, frequency, initial phase, period, time shift, sampling, complex exponentials, Euler's formula, rotating phasors, and phasor addition.

Key idea. Sinusoidal signals are fundamental oscillating building blocks in signals and systems theory; representing real sinusoids via complex exponential phasors transforms complex trigonometric manipulations into simple linear operations in the complex plane.

We begin by introducing a general class of signals commonly called cosine and sine signals. Collectively, they are called sinusoidal signals, or sinusoids. These signals are among the most fundamental in signals-and-systems theory. The mathematical form of a cosine signal is

$$x(t) = A\cos(\omega_0 t + \phi)$$

Here $\cos(\cdot)$ denotes the cosine function. For continuous-time signals, the independent variable t is continuous and represents time. The parameters A , ω_0 , and ϕ are constants for a particular cosine: A is the amplitude, ω_0 is the radian frequency, and ϕ is the initial phase.

For example, consider

$$x(t) = 10\cos(2\pi(440)t - 0.4\pi)$$

In this example $A=10$, $\omega_0=2\pi(440)$ rad/s, and $\phi=-0.4\pi$ rad. The signal oscillates between $+A$ and $-A$ and repeats every $1/440$ second, approximately 2.27 ms.

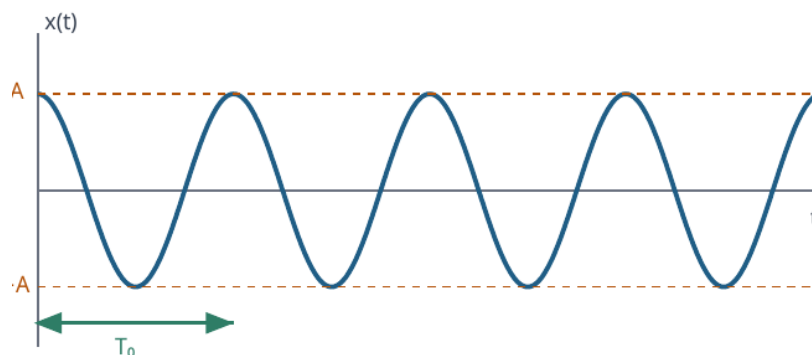


Figure 2.1. Example of a sinusoidal signal: $x(t) = 10\cos(2\pi(440)t - 0.4\pi)$

2.1 An experiment with a tuning-fork

One reason cosine waves are so important is that many physical systems generate signals that can be modelled as sine or cosine functions of time. A tuning fork, after being struck, emits an approximately pure tone. The tone contains essentially one frequency, which is usually marked on the fork. A common tuning fork is “A-440”, because 440 Hz is the frequency of the A above middle C and is widely used for tuning musical instruments.

Experiment: Record the sound of a tuning fork with a microphone connected to the computer’s analogue-to-digital converter, and display the resulting waveform.

2.2 Review of sine and cosine functions

Sinusoidal functions are defined in terms of sine and cosine. In a right triangle, for an angle θ ,

$$\sin\theta = \frac{y}{r}, \quad \cos\theta = \frac{x}{r}$$

Essential trigonometric properties and calculus relations used in sinusoidal analysis include:

- Phase relations: $\sin(\theta) = \cos(\theta - \pi/2)$, $\cos(\theta) = \sin(\theta + \pi/2)$
- Symmetry: $\cos(-\theta) = \cos(\theta)$, $\sin(-\theta) = -\sin(\theta)$
- Pythagorean identity: $\sin^2(\theta) + \cos^2(\theta) = 1$
- Angle sum/difference: $\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$
- Double angle formulas: $\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$, $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$
- Derivatives: $d/d\theta [\sin(\theta)] = \cos(\theta)$, $d/d\theta [\cos(\theta)] = -\sin(\theta)$

2.3 Sinusoidal signals

The most general continuous-time sinusoid is obtained by making the argument of the cosine a linear function of time:

$$x(t) = A\cos(\omega_0 t + \phi) = A\cos(2\pi f_0 t + \phi)$$

A is the amplitude, ϕ is the initial phase in radians, ω_0 is the frequency in radians per second, and f_0 is the frequency in hertz.

Worked example. From the formula below, read the amplitude and frequency directly; the period is $2\pi/\omega_0$ and a peak occurs when $\omega_0 t + \phi = 0$.

$$x(t) = 5\cos(0.3\pi t + 1.2\pi)$$

$$A = 5, \quad \omega_0 = 0.3\pi, \quad f_0 = 0.15 \text{ Hz}, \quad T_0 = \frac{20}{3} \text{ s}, \quad t_m = -4 \text{ s}$$

2.3.1 Relationship between frequency and period

The period T_0 is the duration of one cycle. Periodicity requires

$$x(t + T_0) = x(t)$$

$$A\cos[\omega_0(t + T_0) + \phi] = A\cos(\omega_0 t + \phi)$$

$$\omega_0 T_0 = 2\pi$$

$$T_0 = \frac{2\pi}{\omega_0}$$

$$T_0 = \frac{1}{f_0}, \quad f_0 = \frac{1}{T_0}$$

Therefore, f_0 is the number of cycles per second, measured in hertz.

2.3.2 Relation of phase shift to time shift

The initial phase, together with the frequency, determines the temporal locations of the maxima and minima of a cosine wave. To introduce the idea of a time shift, define the triangular function

$$s(t) = \begin{cases} t, & 0 \leq t \leq 1, \\ \frac{1}{2}(3 - t), & 1 \leq t \leq 3, \\ 0, & \text{otherwise,} \end{cases}$$

Now define $x_1(t)=s(t-2)$. It is nonzero when $0 \leq t-2 \leq 3$, that is, for $2 \leq t \leq 5$. Within this interval,

$$x_1(t) = \begin{cases} t-2, & 2 \leq t \leq 3, \\ \frac{1}{2}(5-t), & 3 \leq t \leq 5, \\ 0, & \text{otherwise,} \end{cases}$$

Thus $x_1(t)$ is simply $s(t)$ shifted two seconds to the right.

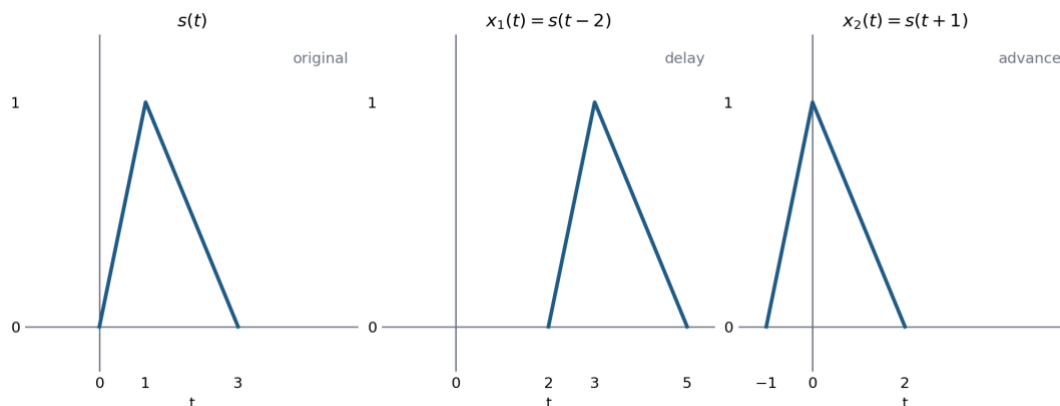


Figure 2.2. Illustration of time shifting: the original signal, a delay, and an advance.

Whenever a signal can be written as $x_1(t)=s(t-t_1)$, it is a time-shifted version of $s(t)$. If $t_1>0$, the shift is to the right and the signal is delayed. If $t_1<0$, the shift is to the left and the signal is advanced.

For a zero-phase cosine $x_0(t)=A \cos(\omega_0 t)$, a time delay can be written as an initial phase shift:

$$x_0(t-t_1) = A \cos[\omega_0(t-t_1)] = A \cos(\omega_0 t + \phi)$$

$$\phi = -\omega_0 t_1$$

$$t_1 = -\frac{\phi}{\omega_0} = -\frac{\phi}{2\pi f_0}$$

$$\phi = -2\pi f_0 t_1 = -2\pi \left(\frac{t_1}{T_0} \right)$$

A positive delay gives a negative phase shift. Because adding any integer multiple of 2π to the cosine argument does not change its value, phase is inherently ambiguous modulo 2π . A common convention is $-\pi < \phi \leq \pi$.

2.4 Sampling and plotting sinusoids

To plot the continuous-time sinusoid

$$x(t) = 20 \cos(2\pi(40)t - 0.4\pi)$$

we evaluate it at discrete times $t_n = nT_s$, where n is an integer. This gives the sample sequence

$$x(nT_s) = 20 \cos(80\pi nT_s - 0.4\pi)$$

where T_s is the sampling period. A Python implementation is:

```
import numpy as np
import matplotlib.pyplot as plt

n = np.arange(-7, 6)
Ts = 0.005
tn = n * Ts
xn = 20 * np.cos(80 * np.pi * tn - 0.4 * np.pi)
```

```
plt.plot(tn, xn)
plt.show()
```

As the sampling period decreases, more samples are taken during each sinusoidal cycle, and the plotted curve appears smoother.

2.5 Complex exponentials and phasors

The analysis and manipulation of sinusoidal signals is greatly simplified by using complex exponentials.

2.5.1 Review of complex numbers

A complex number may be written in Cartesian form as

$$z = x + jy, \quad x = \operatorname{Re}\{z\}, \quad y = \operatorname{Im}\{z\}$$

It may also be represented in polar form by its magnitude r and angle θ :

$$z = r \angle \theta$$

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}, \quad \theta = \operatorname{atan2}(y, x)$$

Euler's formula gives a particularly useful polar notation:

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$z = r e^{j\theta} = r \cos \theta + j r \sin \theta$$

The exponential polar form is especially convenient for complex multiplication and division because magnitudes multiply or divide while angles add or subtract.

2.5.2 Complex exponential signals

A complex exponential signal is defined as

$$\bar{x}(t) = A e^{j(\omega_0 t + \phi)}$$

$$|\bar{x}(t)| = A, \quad \arg\{\bar{x}(t)\} = \omega_0 t + \phi$$

$$\bar{x}(t) = A \cos(\omega_0 t + \phi) + j A \sin(\omega_0 t + \phi)$$

A real cosine is the real part of a complex exponential:

$$x(t) = \operatorname{Re}\{A e^{j(\omega_0 t + \phi)}\} = A \cos(\omega_0 t + \phi)$$

This representation replaces many trigonometric manipulations with simpler algebraic operations on complex exponentials.

2.5.3 The rotating phasor interpretation

For two complex numbers $z_1 = r_1 e^{j\theta_1}$ and $z_2 = r_2 e^{j\theta_2}$, multiplication gives

$$z_3 = z_1 z_2 = r_1 r_2 e^{j(\theta_1 + \theta_2)}$$

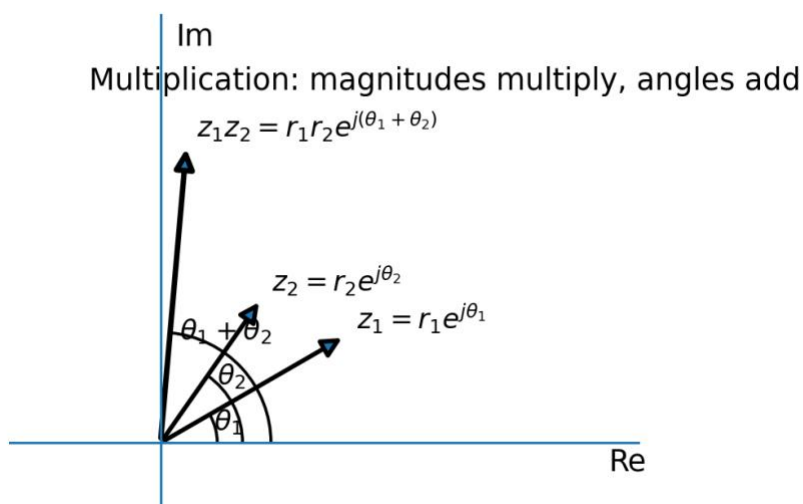


Figure 2.3. Geometric interpretation of complex multiplication.

Define the complex amplitude, or phasor,

$$X = Ae^{j\phi}$$

Then the complex exponential may be written as

$$\tilde{x}(t) = X e^{j\omega_0 t} = A e^{j\phi} e^{j\omega_0 t} = A e^{j(\omega_0 t + \phi)}$$

$$\theta(t) = \omega_0 t + \phi$$

Multiplication by $e^{j\omega_0 t}$ rotates the phasor X . For positive ω_0 , the rotation is counter-clockwise; for negative ω_0 , it is clockwise. One full revolution occurs when the angle increases by 2π , giving

$$\omega_0 T_0 = (2\pi f_0) T_0 = 2\pi \Rightarrow T_0 = \frac{1}{f_0}$$

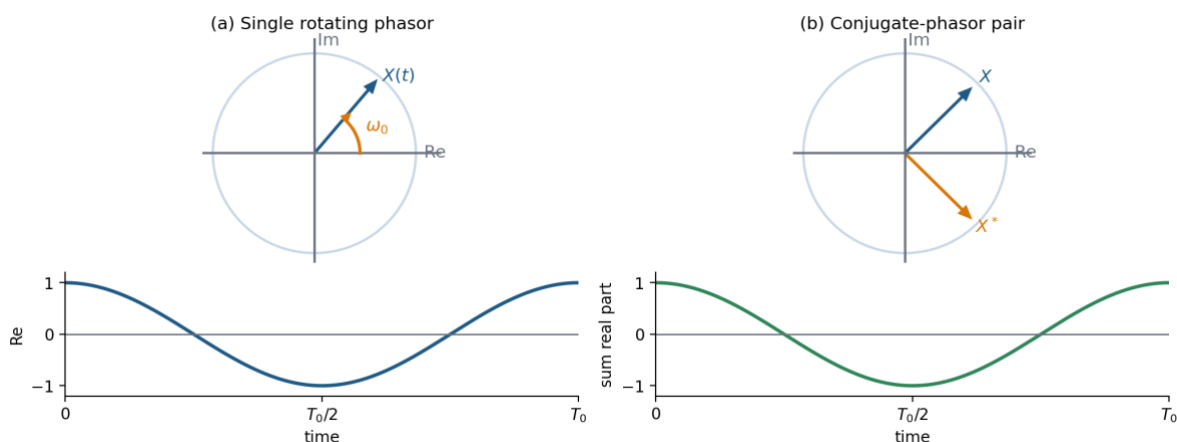


Figure 2.4. Rotating phasors: a single rotating phasor and a conjugate-phasor pair.

2.5.4 Inverse Euler formulas

The inverse Euler formulas express sine and cosine in terms of complex exponentials:

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

Applying the cosine formula to a sinusoidal signal gives

$$\begin{aligned} A\cos(\omega_0 t + \phi) &= \frac{A}{2} [e^{j(\omega_0 t + \phi)} + e^{-j(\omega_0 t + \phi)}] \\ &= \frac{1}{2} X e^{j\omega_0 t} + \frac{1}{2} X^* e^{-j\omega_0 t} \\ &= \frac{1}{2} \bar{x}(t) + \frac{1}{2} \bar{x}^*(t) = \text{Re}\{\bar{x}(t)\} \end{aligned}$$

Here * denotes complex conjugation. A real cosine can therefore be represented by the sum of two complex rotating phasors that are conjugates of one another, corresponding to positive and negative frequencies.

2.6 Phasor addition

A sum of N cosine signals with different amplitudes and initial phases, but with the same frequency, can be reduced to one cosine at that frequency:

$$\sum_{k=1}^N A_k \cos(\omega_0 t + \phi_k) = A \cos(\omega_0 t + \phi)$$

A direct trigonometric proof begins with

$$A_k \cos(\omega_0 t + \phi_k) = A_k \cos \phi_k \cos(\omega_0 t) - A_k \sin \phi_k \sin(\omega_0 t)$$

The phasor approach is simpler.

2.6.1 Addition of complex numbers

Complex numbers add vectorially in the complex plane. Their real parts add, and their imaginary parts add:

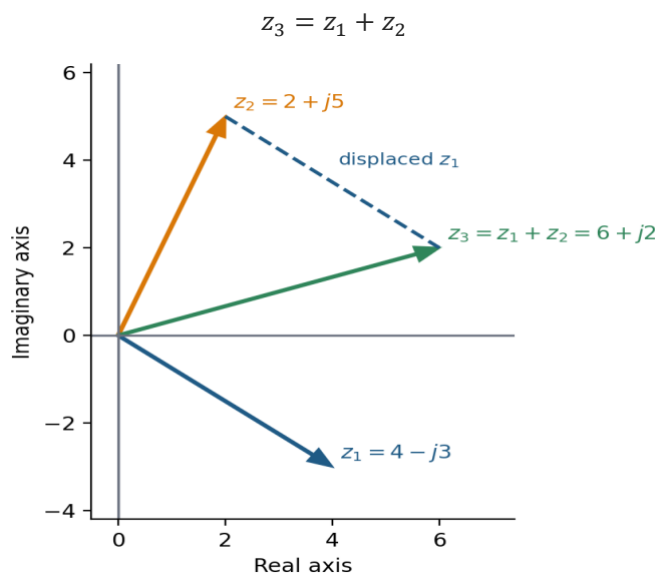


Figure 2.5. Graphical construction of the sum of complex numbers $z_3 = z_1 + z_2$.

2.6.2 Phasor addition rule

The sinusoidal addition rule becomes the following phasor addition:

$$\sum_{k=1}^N A_k e^{j\phi_k} = A e^{j\phi}$$

The proof uses

$$A \cos(\omega_0 t + \phi) = \operatorname{Re}\{A e^{j(\omega_0 t + \phi)}\} = \operatorname{Re}\{A e^{j\phi} e^{j\omega_0 t}\}$$

$$\operatorname{Re}\left\{\sum_{k=1}^N X_k\right\} = \sum_{k=1}^N \operatorname{Re}\{X_k\}$$

Therefore,

$$\begin{aligned} & \sum_{k=1}^N A_k \cos(\omega_0 t + \phi_k) \\ &= \sum_{k=1}^N \operatorname{Re}\{A_k e^{j(\omega_0 t + \phi_k)}\} \\ &= \operatorname{Re}\left\{\left[\sum_{k=1}^N A_k e^{j\phi_k}\right] e^{j\omega_0 t}\right\} \\ &= \operatorname{Re}\{A e^{j\phi} e^{j\omega_0 t}\} \\ &= \operatorname{Re}\{A e^{j(\omega_0 t + \phi)}\} \\ &= A \cos(\omega_0 t + \phi) \end{aligned}$$

Hence, to combine same-frequency sinusoids, convert each sinusoid to its phasor $A_k e^{j\phi_k}$, add the phasors as complex numbers, and convert the resulting phasor $A e^{j\phi}$ back into a cosine.

Numerical example. The two sinusoids below have the same angular frequency. Convert each to a phasor, add the phasors, and return to cosine form.

$$x_1(t) = \cos(77\pi t - \pi), \quad x_2(t) = \sqrt{3} \cos\left(77\pi t + \frac{\pi}{2}\right)$$

$$X_1 = e^{-j\pi} = -1, \quad X_2 = \sqrt{3} e^{j\pi/2} = j\sqrt{3}$$

$$X = X_1 + X_2 = -1 + j\sqrt{3} = 2e^{j2\pi/3}$$

$$x_1(t) + x_2(t) = 2 \cos\left(77\pi t + \frac{2\pi}{3}\right)$$

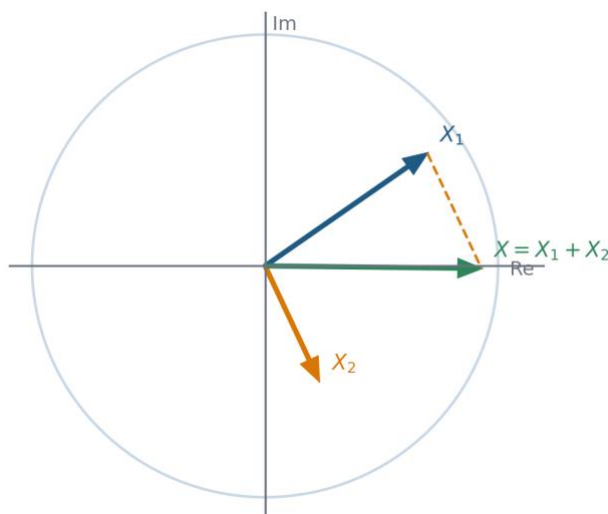


Figure 2.6. Phasor addition for the numerical example.

8. Notation

Symbol	Meaning
A	Sinusoidal amplitude (peak value, $A > 0$)
ω_0	Radian frequency in radians per second (rad/s)
f_0	Cyclic frequency in cycles per second or Hertz (Hz)
ϕ	Initial phase angle in radians
T_0	Fundamental period in seconds ($T_0 = 1/f_0$)
t_1	Time shift / delay in seconds ($t_1 = -\phi/\omega_0$)
T_s	Sampling period in seconds
X	Complex amplitude or phasor ($X = A e^{j\phi}$)
z	Complex number ($z = x + jy = r e^{j\theta}$)
$Re\{\cdot\}, Im\{\cdot\}$	Real part and Imaginary part operators